Spark Ignitor Ignition - Description – General Data

1. General

The spark igniter system consists of an ignition exciter, two spark igniter plugs and shielded cables located on the engine, an igniter selector switch, a common ignition switch and two circuit breakers.

A. Ignition Exciter Unit

The ignition exciter unit, mounted on the engine accessory gear case, transforms the DC input to a pulsed, high voltage output through solid state circuitry, a transformer and diodes. When the unit is energized, a capacitor on the high voltage side of the output transformer is progressively charged until the energy stored is sufficient to ionize a spark gap in the unit and discharge the capacitor across the two spark igniters through a dividing and step-up transformer network. The network also enables the capacitor to discharge automatically in the event of either or both igniters becoming inoperative or input voltage is switched OFF.

B. Ignition Switch

The IGNITION switch is mounted on the overhead console and is marked, MANUAL and NORMAL. When the IGNITION switch is set to NORMAL the ignition circuit is controlled by the starter circuit, so that when the START switch is set to the LEFT or RIGHT position, power is automatically supplied to the selected spark igniters. When the IGNITION switch is set to MANUAL, power is supplied to the spark igniters, independently of the START switch position.

C. Start Switch

The START switch engages the starter of the selected engine and, unless the IGNITION switch is in MANUAL, activates the ignitor of the selected engine (refer to 80-00-00, Starting – General Data).

D. Ignition System Circuit Breakers

Two ignition circuit breakers are incorporated in the system, one to protect each circuit. Both are push-to-reset types marked IGNITION L and IGNITION R and are mounted on the main circuit breaker panel to the rear of the pilot's door. The circuit breakers are 5–amperes.

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